Introduction to Neural Networks

A lab with the TensorFlow Playground

Mehmet Kerem Turkcan

Center for Smart Streetscapes, Columbia University

October 24, 2024

Tool	TensorFlow Playground	Address	playground.tensorflow.org
Computer	Any, with internet access	Browser	Chrome recommended

Introduction

In this lab you will explore the ideas behind neural networks with the TensorFlow Playground, an interactive web application for visualizing and experimenting with different neural network configurations.

Objectives

By the end of this lab, you should be able to:

- understand the basic components of a neural network,
- experiment with different network architectures,
- analyze how hyperparameters affect network performance,
- visualize decision boundaries for different classification problems.

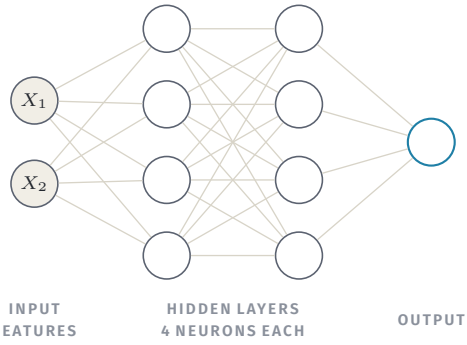

Figure 1 The network you will build in Part 2: two input features, two hidden layers of four neurons, and one output.

Procedure

Part 1 Get to know the TensorFlow Playground

1. Open the TensorFlow Playground in your web browser.
2. Find the main components: the input features (X_1, X_2), the hidden layers and their neurons, the output layer, the data points (orange and blue), and the training loss and test loss.

Part 2 A simple classification problem

1. Set up a simple classification problem:
   - choose **Circle** as the data type,
   - set the noise to 0,
   - use 2 hidden layers with 4 neurons each,
   - set the learning rate to 0.03,
   - use **tanh** as the activation function.
2. Click **Run** and watch how the network learns to classify the points.
3. Record the final training loss and test loss after 1000 epochs.

Part 3 Experiment with hyperparameters

1. Change the following settings one at a time, keeping the others constant:
   (a) the number of hidden layers (1, 2, 3),
   (b) the number of neurons per layer (2, 4, 8),
   (c) the learning rate (0.001, 0.01, 0.1),
   (d) the activation function (ReLU, tanh, sigmoid).
2. For each change, run the network for 1000 epochs and record the final training loss, the final test loss, and how quickly the network converges.

Part 4 A harder problem

1. Choose the **Spiral** data type from the left column.
2. Experiment with different network architectures until you find one that classifies the spiral data successfully.
3. Record your final network configuration and the resulting losses.

Questions for analysis

Q1 How does the number of hidden layers affect the network's ability to learn complex patterns?

Q2 What role does the activation function play in the network's performance?

Q3 How does the learning rate affect the training process?

Q4 For the spiral data, which network configuration worked best? Why do you think this is the case?

Q5 How does adding noise to the data affect the network's performance? Experiment and explain your observations.